

Multivariate Regression

Empirical Methods for Finance

Prof. Virginia Gianinazzi

Nova SBE

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Outline

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- 2 Omitted Variables Bias
- 3 Multivariate Regression Analysis
- 4 Linear regression with dummy variables

Unbiasedness of OLS

Example: Does Pedigree Predict Performance?

$$RET = \beta_0 + \beta_1 SAT + u$$

- We estimate $\hat{\beta}_1 > 0$: managers who attended higher-SAT undergraduate institutions have systematically higher excess returns
- Asking
 - ▶ Does the model identify the causal effect of pedigree on performance?
 - ▶ Does the model identify β_1

is the same as asking

- ▶ Is $\hat{\beta}_1$ coming from estimating this model an **unbiased estimator** of β_1 ?

Uncertainty and sampling variance

- Different samples will generate different estimates ($\hat{\beta}_1$) for the “true” β_1 :
 - ▶ $\hat{\beta}_1$ is a **random variable**, β_1 is a constant
- Being a random variable, $\hat{\beta}_1$ has a sampling distribution, and therefore an expected value $E(\hat{\beta}_1)$ and a variance $Var(\hat{\beta}_1)$

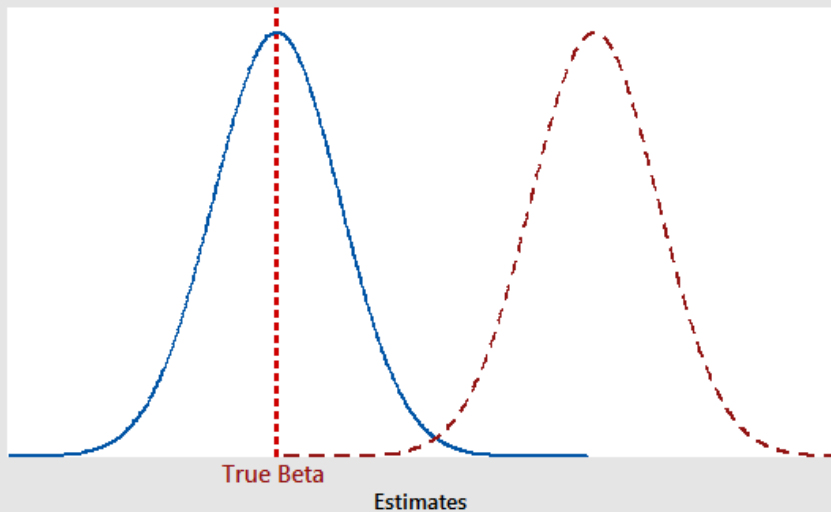
Unbiasedness of OLS

- We say that an estimator is unbiased if

$$E(\hat{\beta}) = \beta$$

- Unbiasedness is the idea that if we could take as many random samples on Y as we want from the population, and compute an estimate each time, the average of these estimates would be equal to the true β

Biased and Unbiased Sampling Distributions of Beta Estimates



We Need to Make Assumptions about the Population Model

- The statistical properties concern the distributions of $\hat{\beta}_0$ and $\hat{\beta}_1$ over different random samples from the population
- These therefore depend not only on how the estimators were constructed, but also on the nature of the population generating the data
- Hence, to establish the statistical properties of the OLS estimators, we need to make **assumptions on the population**

Simple Linear Regression Assumptions

SLR.1 (Linear in Parameters)

The population model can be written as $y = \beta_0 + \beta_1 x + u$

SLR.2 (Random Sampling)

We have a random sample of size N , $\{(x_i, y_i) : i, \dots, N\}$ following the population model.

SLR.3 (Sample Variation in the Explanatory Variable)

The sample outcomes on x_i are not all the same value. In other words, the sample variance of x is not zero.

SLR.4 (Zero Conditional Mean)

In the population, the error term has zero mean given any value of the explanatory variable:

$$E(u|x) = E(u) = 0$$

Unbiasedness of OLS

Under SLR.1 through SLR.4 the OLS estimator is **unbiased** for any sample size n , i.e.

$$E(\hat{\beta}) = \beta$$

(Proof: video)

Omitted Variables Bias

Omitted Variables Bias

- Suppose that RET is determined by

$$RET = \beta_0 + \beta_1 SAT + \beta_2 EXP + u$$

which satisfies SLR.1-4

- However, we estimate the model

$$RET = \beta_0 + \beta_1 SAT + v$$

where $v = \beta_2 EXP + u$

- From estimating this model we get $\tilde{\beta}_1$
- Is $\tilde{\beta}_1$ an unbiased estimator of β_1 ?

Omitted Variables Bias

- $\tilde{\beta}_1$ biased means

$$E(\tilde{\beta}_1) = \beta_1 + \text{BIAS} \neq \beta_1$$

- In which direction is our estimate biased?
- We can derive the bias from omitting *EXP* to be

$$\text{BIAS} = \tilde{\delta}_1 \beta_2$$

where $\tilde{\delta}_1 = \frac{\text{SampleCov}(\text{EXP}, \text{SAT})}{\text{SampleVar}(\text{SAT})}$

Omitted Variables Bias

- From writing $E(\tilde{\beta}_1) = \beta_1 + \tilde{\delta}_1\beta_2$ we see that $\tilde{\beta}_1$ is biased unless

Multivariate Regression Analysis

Multiple Linear Regression

- Very hard to draw ceteris paribus conclusions in univariate framework
- Explicitly control for many other factors that simultaneously affect y :

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + u$$

Multiple Linear Regression

Example: Does Pedigree Predict Performance?

$$RET = \beta_0 + \beta_1 SAT + \beta_2 EXP + u$$

- Interpretation of model parameters:

Multiple Linear Regression

Example: Does Pedigree Predict Performance?

$$\widehat{RET} = -7.5388 + 0.0062SAT$$

$$\widehat{RET} = -5.8348 + 0.0053SAT + 0.1914EXP$$

- Interpretation of the estimated coefficient on SAT from the ... ?
 - ▶ univariate model?
 - ▶ multivariate model?

Multiple Linear Regression

Example: Does Pedigree Predict Performance?

$$\widehat{RET} = -7.5388 + 0.0062SAT$$

$$\widehat{RET} = -5.8348 + 0.0053SAT + 0.1914EXP$$

- What does the change in the coefficient on SAT tell us about omitted variable bias in the univariate model?
- To provide convincing evidence that pedigree matters for performance you need to control for many possible determinants of returns, because if these are correlated with SAT they will bias your results

Linear regression with dummy variables

Dummy Variables

Example: Does Pedigree Predict Performance?

$$RET = \beta_0 + \beta_1 SAT + \beta_2 EXP + \delta_0 MAST + u$$

- How do we incorporate qualitative information, e.g. if the manager has a master degree or not?

Dummy Variables

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$$RET = \beta_0 + \beta_1 SAT + \beta_2 EXP + \delta_0 MAST + u$$

Dummy Variables

Example: Does Pedigree Predict Performance?

$$RET = \beta_0 + \beta_1 SAT + \beta_2 EXP + \delta_0 MAST + u$$

- How do we interpret δ_0 ?

Dummy Variables

Example: Does Pedigree Predict Performance?

$$\widehat{RET} = -5.8269 + 0.0053SAT + 0.1922EXP - 0.0985MAST$$

- Intercept for no master :
- Intercept for master:

ONLY Dummy Variables

Example: Does Pedigree Predict Performance?

$$\widehat{RET} = -0.7238 + 0.3349MAST$$

- Consider now a regression with only the dummy on the RHS
- How do we interpret the coefficients now?

INTERACTIONS With Dummy Variables

Example: Does Pedigree Predict Performance?

$$\widehat{RET} = -0.7238 + 0.3349MAST$$

- Dummies only shift intercept for different groups. What about slope?
- For instance, the quality of the undergrad institution may matter less if you have a master